1.38 Applications of Calculus

Worked Solutions

1 a)

$$\frac{dC}{dx} = 5.8$$

For each additional mile travelled, the cost increases by £5.80

Units: £/mile

1 b)

$$\frac{dh}{dt} = -2t + 6 \quad \text{so} \quad \left.\frac{dh}{dt}\right|_{t=10} = -14$$

At $t = 10$ exactly, the height of the ball is decreasing at a rate of 14 ms^{-1}

Units: ms^{-1}

1 c)

$$\frac{dT}{dt} = 0.03t^2 - 0.4t \quad \text{so} \quad \left.\frac{dT}{dt}\right|_{t=5} = -1.25$$

At $t = 5$ exactly, the temperature of the substance is decreasing at a rate of 1.25°C per hour.

Units: °C per hour

2 a)

$$\frac{dN}{dt} = -0.98t + 10$$

$$\left.\frac{dN}{dt}\right|_{t=5} = 5.1$$

On day 5 there is 5.1 people per day gaining the virus

2 b)

$$\left.\frac{dN}{dt}\right|_{t=15} = -4.7$$

On day 15 there is 4.7 fewer people per day gaining the virus

2 c)

Max pt when $\frac{dN}{dt} = 0 \ \therefore -0.98t + 10 = 0$

$$(10.2, 54.6)$$

2 d)

The model says that at $t = 10.2$ days, the maximum amount of people with the virus is 54.6 people.

3 a)

$$\frac{dh}{dn} = -0.056n + 1.95$$

It represents the rate in change (per day) of the height of the seedling (in mm)

3 b)

Max when $\frac{dh}{dn} = 0$

$$-0.056n + 1.95 = 0 \ \therefore n = 34.8$$

So according to the model during the 34th day it reaches its maximum height of 40.2 mm

3 c)

The model is no longer valid after reaching its maximum because it then models the height as decreasing.

4 a)

$$\frac{dv}{dt} = 7t^2 - 16t + 4$$

Stat pt when $\frac{dv}{dt} = 0$ so $7t^2 - 16t + 4 = 0$

$$\therefore t = 2, \ t = \frac{2}{7}$$

$$(2, 4.67), (0.286, 10.5)$$

4 b)

When $t = 0$, $v = 10$

When $t = 3$, $v = 13$

Therefore when $t = 2$ seconds the object has its lowest velocity, and when $t = 0.286$ seconds, the object has a local maximum velocity (the actual maximum is 13 ms^{-1})

5 a)

$$\frac{dy}{dt} = 3t^2 - 16t + 15$$

Stat pt when $\frac{dy}{dt} = 0$

$$3t^2 - 16t + 15 = 0 \ \Rightarrow \ t = \frac{8}{3} \pm \frac{1}{3}\sqrt{19}$$

When $t = \frac{8}{3} - \frac{1}{3}\sqrt{19} = 1.21$ s, $y = 56.2$ cm

When $t = 5$ s, $y = 48$ cm

$\therefore$ maximum depth of water is 56.2 cm

5 b)

$$3t^2 - 16t + 15$$

Min when $6t - 16 = 0$ so $t = \frac{8}{3} = 2.67$ seconds

$$\left.\frac{dy}{dt}\right|_{t=\frac{8}{3}} = -\frac{19}{3} = -6.33 \quad (3 \text{ SF})$$

$$y = 50.1 \text{ cm}$$

5 c)

The boat is moving downwards as this point happens after the local maximum.

6 a)

$$h'(x) = 6x^2 - 60x + 94$$
$$h'(0) = 94$$

6 b)

Initially, the gradient says that for every 100 feet the height of the rollercoaster increases by 94 feet

$$\theta = \tan^{-1}\left(\frac{94}{100}\right) = 43.2^\circ$$

6 c)

Max and min heights when $h'(x) = 0$

So $6x^2 - 60x + 94 = 0$ so $x = 5 \pm \frac{2}{3}\sqrt{21}$

Max when $x = 1.944949537\ldots$

The maximum height occurs 194 feet horizontally from the beginning of the rollercoaster and reaches a height of 234 feet above the ground.

7)

$$\frac{1}{2}\int_0^2 -10x^3 + 14x^2 + \frac{3}{2}x\,dx = \left[-\frac{5}{4}x^4 + \frac{7}{3}x^3 + \frac{3}{8}x^2\right]_0^2$$
$$= \left(-20 + \frac{56}{3} + \frac{3}{2}\right) - (0) = \frac{1}{6} = 0.167 \text{ metres}$$

8 a)

£1274

8 b)

$$\frac{1}{12}\int_0^{12} 0.5t^3 + 3.5t^2 - 12t + 50\,dt$$
$$= \left[\frac{1}{96}t^4 + \frac{7}{72}t^3 - \frac{1}{2}t^2 + \frac{25}{6}t\right]_0^{12}$$
$$= (216 + 168 - 72 + 50) - (0)$$
$$= 362$$

So £362